

Earl Chirchir Graduate Mechatronic Engineer

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LinkedIn: <https://linkedin.com/in/earl-chirchir> | GitHub: <https://github.com/earlchirchir>

Graduate Visa (Full Time Right to Work) | Portfolio: earlchirchir.github.io

Summary

Mechatronic Engineer (MSc, Distinction) with 3+ years experience in the design, development and integration of automation and embedded systems within industrial and manufacturing environments. Strong technical capability in robotics and trajectory planning, control systems, embedded programming and CAD/CAM design, with proven experience delivering solutions from concept and simulation through to implementation. A collaborative engineer with a practical, problem-solving mindset, committed to developing reliable, scalable and safety-conscious automation solutions aligned with modern industrial and sustainability standards.

Skills

- **Robotics:** Trajectory Planning for cobots (MATLAB, RoboDK)
- **Computer Aided Design:** NX Siemens, Autodesk Inventor, SolidWorks
- **Programming:** MATLAB, C++ (Arduino/ESP32), Python
- **Circuit Design & PCB fabrication:** Fritzing, Proteus, KiCAD, AutoDesk Eagle
- **Embedded Programming:** Arduino & ESP32 frameworks, Electronic sensors and actuators
- **Good communication skills:** Pearson Test for English (77 points), IELTS (8.0/9.0)

Key Achievements

- **Optimised robotic trajectory planning** for an ABB CRB 15000 collaborative robot using a Bang-Coast-Bang time law, delivering smoother motion and improved cycle efficiency through MATLAB simulation and RoboDK validation.
- **Designed and deployed an industrial IoT data logging and telemetry system** for industrial machinery, increasing remote data accuracy by ~40% and enabling real-time monitoring via SQL databases and Power BI dashboards.
- **Developed and implemented industrial HMIs** using Nextion displays for weighing and production systems, improving operator interaction, interface reliability, and overall machine uptime.
- **Engineered an end-to-end embedded and data integration architecture**, linking microcontrollers, remote servers and analytics platforms using JSON-based communication for real-time industrial insights.

Work Experience

Development Engineer, OJE Engineering

11/2022 – 12/2024 | Kenya

- Led the maintenance of an IoT-based device for remote data logging for tea-picking machinery for 2 industries, enhancing real time transmission of tea data.
- Designed and implemented a Human-Machine Interface (HMI) for industrial weighing bins using Nextion displays to improve operator-machine communication and work flow efficiency, increasing machine operational uptime through improved interface reliability.
- Supervised the configuration of 2 remote web servers and SQL databases using Postman to enable secure Internet of Things (IoT) data handling.
- Programmed and debugged embedded systems (Arduino Nano) for industrial automation, ensuring reliable performance.
- Engineered a JSON-based data interchange system to connect microcontrollers, servers and Power BI for real-time analytics.

Enforcement and Consumer Protection Engineer

01/2019 – 12/2019 | Kenya

Energy and Petroleum Regulatory Authority (EPRA)

- Supported regulatory operations in the petroleum sector, with emphasis on licensing, construction permit issuance, and fuel price monitoring in line with statutory requirements.
- Contributed to regulation of the electric power sub-sector, including licensing of electricians, registration of electrical contractors, and investigation of disputes in accordance with the Energy Act.
- Reviewed and approved electricity meters to ensure accuracy and compliance for measurement of electrical energy consumption.
- Performed nationwide petroleum price surveillance, monitoring compliance with maximum regulated retail prices and inspecting pipeline and storage facilities.
- Conducted fuel quality monitoring and testing to prevent motor fuel adulteration and illegal diversion or dumping of export-bound fuels.

Systems Engineer Intern, Fab Lab Nairobi Maker Space

07/2017 – 09/2017 | Kenya

- Developed a microcontroller-based calculator using Arduino and C++.
- Designed PCBs using KiCAD and AutoDesk Eagle.
- Soldered PCBs meant to create a speed governor prototype.

Industrial Maintenance Intern, James Finlay (Kenya) Limited

05/2016 – 07/2016 | Kenya

- Implemented predictive diagnostics and calibrated process control instruments, improving overall maintenance efficiency for boiler, compressor, and conveyor systems.
- Ensured production continuity and enhanced machine availability through comprehensive preventive maintenance programs.
- Reduced breakdown incidents by proactively identifying and addressing potential failures.

Additional Work Experience

Health Care Assistant, Omni Healthcare • Support service users with personal care, mobility, and daily routines. • Promote independence, dignity, and wellbeing of service users. • Support service users with non-clinical assistance and companionship. • Communicate effectively with healthcare professionals and team colleagues. • Support social and recreational activities of service users to improve morale. • Maintain accurate care records and handover notes.	04/2026 – Present Swansea, Wales
Production Operative, New England Seafood International (NESI) • Worked in a regulated, high-volume production environment, following strict safety, hygiene, and quality procedures aligned with industrial best practices. • Supported production line efficiency through coordinated operations, inspections, and adherence to standard operating procedures.	10/2025 – 12/2025 London, UK
Kitchen Assistant, Get Golfing CIO (Hampton Court Palace Golf Club) • Supported kitchen operations in a high-volume hospitality environment, ensuring compliance with health, safety, and food hygiene standards.	08/2024 – 10/2025 London, UK
Hearing Healthcare Intern, Starkey Hearing Foundation • Conducted community hearing assessments and supported clinical fitting of hearing aids, assisting in screening over 100+ within 2 months. • Shadowed audiologists during outreach missions, ensuring proper technical setup of diagnostic equipment.	02/2020 – 03/2020 Kenya

Projects

MSc Dissertation: Trajectory Planning for an ABB CRB 15000 Robot Using Bang-Coast-Bang Time Law  • Optimised robotic trajectory planning for ABB CRB 15000 robot using Bang-Coast-Bang time law reducing cycle time improving motion smoothness in simulation. • Technologies: MATLAB, RoboDK	05/2024 – 02/2025
Onboard Monitoring System  • Developed and implemented an IoT-based telemetry data logging solution for tea-picking machinery, enhancing remote data accuracy by 40%. • Technologies: Arduino, SQL, PowerBI, PCBs, Soldering	11/2022 – 04/2023
Design and fabrication of an Automated Shopping Cart  • Designed and implemented an autonomous navigation system. • Technologies: Sensors, Actuators, Arduino, PCBs, Soldering	11/2018
Design and fabrication of an Automated Chicken Coop  • Designed IoT enabled feeding, watering, and egg collection modules. • Technologies: Arduino, PCBs, Soldering	01/2018

Education

MSc Mechatronic Systems, Kingston University	01/2024 – 02/2025 United Kingdom
Classification : Distinction. Key Modules : Engineering Research Techniques, Entrepreneurship and Quality Management (85%), Individual Project (81%), Control Systems with Embedded Implementation (79%), Mechatronic Design and Automation (63%). Dissertation : Trajectory Planning for an ABB CRB 15000 Robot Using Bang-Coast-Bang Time Law.	
BSc Mechatronic Engineering, Upper Second Class Honors (2:1) <i>Jomo Kenyatta University of Agriculture and Technology</i> Key Modules: Automation and Robotics, Instrumentation and Control, Thermodynamics, Microprocessors, and Mechatronic Systems Design. Leadership: Co-led Engineering Students in the Innovation Club	09/2013 – 11/2018 Kenya

Awards

Aspiring Innovator Intervarsity Competition, Riara University First Place, Biashara Bunifu (Innovative Business) Category: Design and fabrication of an Automated Chicken Coop	01/2018
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Certifications

- Data Science: Moringa School
- HTML5 & CSS3: Pirple Inc.
- Python Certificate Course: Programming Hub
- Autodesk Inventor Essential Training: LinkedIn Learning